Jiaqi Wang

PhD candidate in Astrophysics · expected June 2027
Tsung-Dao Lee Institute & School of Physics and Astronomy, Shanghai Jiao Tong University
Visiting researcher, Institute for Computational Cosmology, Durham University (Jul 2024–Jun 2026)
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RESEARCH PROFILE

Cosmologist working on large-scale structure with simulations, weak gravitational lensing and statistical inference. Current focus: interpretable summary statistics of the nonlinear cosmic web and their robustness to baryonic feedback. Introduced *persistence strips*, which roughly double the neutrino-mass constraining power of unbinned Betti curves (0.05 eV from the total-matter field of a $(350\,\mathrm{Mpc})^3$ simulation volume) [1]; earlier work measured weak-lensing halo masses and halo mass functions for DESI Legacy Imaging Surveys groups [2].

Interests neutrino mass and dark energy from large-scale structure · topological and beyond-two-point statistics · weak-lensing cosmology · simulation-based inference · interpretable machine learning for cosmology

EDUCATION

Shanghai Jiao Tong University Tsung-Dao Lee Institute & School of Physics and Astronomy — 2022 – present
PhD in Astrophysics (in progress), expected June 2027 · Tsung-Dao Lee PhD Honor Program
Supervisors: Xiaohu Yang and Carlos S. Frenk.
Thesis (working title): *Probing Cosmology with Cosmic-Web Topology and Weak Gravitational Lensing*.

Durham University Institute for Computational Cosmology — Jul 2024 – Jun 2026
Visiting researcher (two-year CSC-funded doctoral research exchange) · Collaborators: Carlos S. Frenk, Shaun Cole, Willem Elbers.

Shanghai Jiao Tong University School of Physics and Astronomy — 2018 – 2022
BSc in Astrophysics · Minor in Philosophy · Outstanding Undergraduate Thesis (university-level award).

RESEARCH EXPERIENCE

Neutrino mass from the topology of the cosmic web — 2024 – present

Durham ICC · with W. Elbers, C. S. Frenk and S. Cole · first-author paper [1]

- **Question** Can the multiscale topology of the cosmic web isolate the weak neutrino-mass signal that conventional probes leave degenerate with other cosmological parameters?
- **Contribution** Introduced *persistence strips*, which segment Betti curves by topological persistence and condense 2D persistence diagrams into a set of 1D curves. Built the full analysis: statistics measured on FLAMINGO simulations, a Gaussian-process emulator over a 10-parameter $w_0w_a\mathrm{CDM}+\nu$ space, held-out validation, covariance modelling and Bayesian parameter inference.
- **Result** Neutrino-mass uncertainty of 0.05 eV from the total-matter field (0.13 eV for dark matter only) in a $(350\,\mathrm{Mpc})^3$ volume, about twice the constraining power of unbinned Betti curves; void topology is the most constraining.
- **Physical origin** Controlled information-removal tests trace the signal to two comparably important sources: the neutrino mass fraction (largest in underdense regions) and the neutrino-induced change in the dark-matter distribution.
- **Ongoing** Testing which topological signatures remain robust to baryonic feedback, using FLAMINGO hydrodynamical simulations; manuscript in preparation.

Weak-lensing halo masses for galaxy groups and clusters — 2021 – present

Shanghai Jiao Tong University · first-author paper [2]; lensing analyses in [5, 6]

- **First-author study** Measured and modelled group–galaxy weak-lensing profiles of DESI Legacy Imaging Surveys DR9 groups and clusters at $0.1 \le z < 0.6$ (5 redshift × 7 mass bins), using the DECaLS Fourier_Quad shear catalogue and a model with off-centring. Halo masses, concentrations and bias were self-consistent across three group-centre definitions, giving cumulative halo mass functions to $z = 0.6$ with 0.03–0.09 dex mass precision.
- **Collaborative analyses** Primary responsibility for the galaxy–galaxy lensing measurements and profile fitting in the first observational measurement of the characteristic depletion radius of dark-matter haloes [6] and in a comparison of red-sequence and halo-based optical cluster finders [5].

- **DESI collaborations** Provided gravitational-lensing measurements for early tests of the theoretical feasibility of Fundamental-Plane Lensing [3]. Contributed to group-catalogue preparation and scientific discussions for the assembly-bias study [4], alongside related ongoing work using a different methodology.
- **Shear measurement** Training project with the Fourier_Quad method, testing metacalibration and PSF-interpolation strategies for undersampled images.

HONOURS & FUNDING

2025 Outstanding Presentation Award (Second Prize), First Jiao-Chuang Doctoral Academic Forum
2022 Outstanding Undergraduate Thesis, Shanghai Jiao Tong University (university-level award).
2024 China Scholarship Council (CSC) scholarship for doctoral research exchange at Durham University.
2022 Shanghai Outstanding Graduate.
2020 National Scholarship; National Second Prize, China Undergraduate Mathematical Contest in Modeling.
2019 Annual Scholarship, National Astronomical Observatories, Chinese Academy of Sciences

PUBLICATIONS

9 refereed papers (2 first-author) and 3 submitted · complete record: ORCID 0000-0003-0086-790X

First author

[1] **Wang, J.**, Elbers, W., Frenk, C. S., Cole, S., Yang, X., McCarthy, I. G., & van de Weygaert, R. (2026). "Revealing the neutrino mass through persistent homology of the cosmic web." *MNRAS*, 549, stag937. doi:10.1093/mnras/stag937

[2] **Wang, J.**, Yang, X., Zhang, J., et al. (2022). "Halo Properties and Mass Functions of Groups/Clusters from the DESI Legacy Imaging Surveys DR9." *ApJ*, 936, 161. doi:10.3847/1538-4357/ac8986

Major contributions

[3] Xu, K., **Wang, J.**, Sheth, R. K., et al. (2026). "Sizing the Universe with DESI Galaxy Sizes: Plain Fundamentals of Fundamental-Plane Lensing." DESI collaboration paper (47 authors), submitted. arXiv:2609.09151
Role: gravitational-lensing measurements for early tests of theoretical feasibility

[4] Shao, Z., Zu, Y., Salcedo, A. N., **Wang, J.**, et al. (2025). "Direct Measurement of Galaxy Assembly Bias using DESI DR1 Data." DESI collaboration paper (57 authors), submitted. arXiv:2510.20896
Role: group-catalogue preparation and scientific discussions

[5] Golden-Marx, J. B., Zu, Y., **Wang, J.**, Li, H., Zhang, J., & Yang, X. (2023). "Satellite content and halo mass of galaxy clusters: comparison between red-sequence and halo-based optical cluster finders." *MNRAS*, 524, 4455. doi:10.1093/mnras/stad2174
Role: galaxy–galaxy lensing measurements and profile fitting

[6] Fong, M., Han, J., Zhang, J., Yang, X., Gao, H., **Wang, J.**, et al. (2022). "First measurement of the characteristic depletion radius of dark matter haloes from weak lensing." *MNRAS*, 513, 4754. doi:10.1093/mnras/stac1263
Role: galaxy–galaxy lensing measurements and profile fitting

Other co-authored papers

[7] Li, Q., Yang, X., Hong, W., Shi, F., Zhang, Y., **Wang, J.**, et al. (2026). "ELUCID-DESI II. Revealing dark matter mass, tidal, and velocity (MTV) fields using galaxy group phase information." Submitted to *ApJ*. arXiv:2608.26668

[8] Zheng, Y.-L., Yang, X., Liu, T., et al. incl. **Wang, J.** (2026). "Unveiling the Sources of X-Ray Luminosity in DESI Galaxy Groups: Insights from the SRG/eROSITA All-sky Survey." *ApJ*, 997, 111. doi:10.3847/1538-4357/ae21d5

[9] Li, Q., Kilbinger, M., Luo, W., et al. incl. **Wang, J.** (2024). "Black Hole–Halo Mass Relation from UNIONS Weak Lensing." *ApJL*, 969, L25. doi:10.3847/2041-8213/ad58b0

[10] Wang, Y., Yang, X., Gu, Y., et al. incl. **Wang, J.** (2024). "Measuring the Conditional Luminosity and Stellar Mass Functions of Galaxies by Combining the Dark Energy Spectroscopic Instrument Legacy Imaging Surveys Data Release 9, Survey Validation 3, and Year 1 Data." *ApJ*, 971, 119. doi:10.3847/1538-4357/ad5294

[11] Gu, Y., Yang, X., Han, J., et al. incl. **Wang, J.** (2024). "CSST large-scale structure analysis pipeline: I. Constructing reference mock galaxy redshift surveys." *MNRAS*, 529, 4015. doi:10.1093/mnras/stae762

[12] Wang, K., Mo, H. J., Chen, Y., Wang, H., Yang, X., **Wang, J.**, Peng, Y., & Cai, Z. (2024). "Characterizing the assembly of dark matter haloes with protohalo size histories: I. Redshift evolution, relation to descendant haloes, and halo assembly bias." *MNRAS*, 528, 2046. doi:10.1093/mnras/stae163

TALKS & POSTERS

Contributed talks

Jul 2026 European Astronomical Society Annual Meeting (EAS 2026), Lausanne, Switzerland
Jul 2026 2026 Annual Science Meeting of the Chinese Space Station Telescope (CSST), Kashgar, China

Apr 2026 Euclid UK Annual Meeting, University of Manchester, UK
Jan 2026 DEX-XXII, Durham–Edinburgh eXtragalactic Workshop: "The Universe at All Scales", Edinburgh, UK
Nov 2025 3rd Shanghai Assembly on Cosmology and Structure Formation, Tsung-Dao Lee Institute, Shanghai
Nov 2024 NFW@30, Zhangjiajie, China

Posters and short talks

Aug 2026 Poster, New Frontiers in Cosmology 2026, A Coruña, Spain
Jul 2025 Three-minute public-facing talk, UK National Astronomy Meeting (NAM 2025), Durham, UK
Jun 2025 Poster, AI Applications in Data-Intensive Science, University College London, London, UK

Departmental seminars

2025–26 Durham FLAT seminar series (Friday Lunchtime Astrophysics Talks), ICC & CEA, Durham University (Feb 2025, May 2026)

METHODS & COMPUTING

Statistical inference	Bayesian parameter inference and MCMC; Gaussian-process emulation trained on simulation suites; covariance estimation; validation on held-out simulations; uncertainty propagation.
Topology	Persistent homology of cosmic-web fields (persistence diagrams, Betti curves, persistence strips); multiscale, environment-dependent statistics.
Weak lensing	Galaxy–galaxy and group–galaxy lensing measurements; halo-profile modelling (mass, concentration, bias, off-centring); shear measurement with Fourier_Quad.
Simulations & data	FLAMINGO simulations; DESI Legacy Imaging Surveys DR9 / DESI DR3 group catalogue; DECaLS Fourier_Quad shear catalogue.
Programming	Python (NumPy, SciPy, Pandas), C/Cython and Fortran; Linux and Git.
HPC	SLURM; multiprocessing and multithreading; shared-memory processing, chunked data reads and I/O optimisation; experience with CUDA/GPU computing environments.
Machine learning	Gaussian-process emulation; AI-agent workflows, tool use and output validation.

PUBLIC ENGAGEMENT

2020–23 Public lectures: four talks at three Shanghai high schools (2020–2022) and "A Guide to Stargazing in the City" at the Shanghai Astronomy Museum (2023).
2024 Demonstration assistant, Durham University outreach: solar-telescope and "Universe Machine" demonstrations. In the latter, visitors varied cosmological parameters to explore simulated universes from a precomputed simulation library, learning about the composition and evolution of the Universe.
2021–26 Contributed to early operations and public engagement of the SJTU Guangqi Observatory.

TEACHING, MENTORING & SERVICE

2022–24 Teaching assistant, University Physics Laboratory and Introduction to Astrophysics, Shanghai Jiao Tong University; guided laboratory experiments and supported student coding work.
2020–21 President, student union, School of Physics and Astronomy, Shanghai Jiao Tong University; established a science-outreach column.

REFERENCES

- **Xiaohu Yang** — Shanghai Jiao Tong University; PhD supervisor. xyang@sjtu.edu.cn
- **Carlos S. Frenk** — Durham University; PhD supervisor. c.s.frenk@durham.ac.uk
- **Shaun Cole** — Durham University; research collaborator. shaun.cole@durham.ac.uk
- **Willem Elbers** — Fellow, Durham University; research collaborator. whe@willemelbers.com